



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
29-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

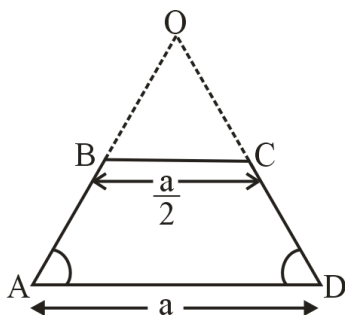
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

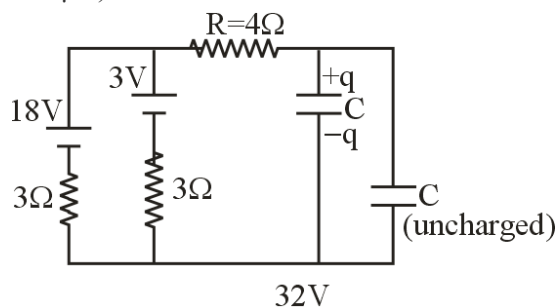
1. Consider a uniform sheet ABCD, which is a part of an equilateral triangular sheet of a side 'a' as shown in the figure. Choose the correct option regarding electric field E at point O due to this sheet.



- (A) Magnitude of E increases with increase in 'a' (keeping charge density same)
- (B) Magnitude of E remains same with increase in a (keeping total charge constant)
- (C) If charge density is σ and $a = 1$ m magnitude of electric field E is equal to $\frac{7\sigma}{44\epsilon_0} \ln \sqrt{2}$.
- (D) If charge density is σ and $a = 2$ m magnitude of electric field E is equal to $\frac{7\sigma}{22\epsilon_0} \ln \sqrt{2}$.

Space for Rough Work

2. An open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured value of length ℓ (in cm) of the pipe and radius r (in cm) of the pipe are $\ell = 94.0 \pm 0.1$, $r = 5.0 \pm 0.05$. The velocity of sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork in this experiment will be.
- (A) 0.16 (B) 0.64 (C) 1.2 (D) 1.6
3. The circuit is connected at $t = 0$ then what is heat loss in $R = 4 \Omega$ resistance till the capacitor gets fully charged. (given $C = 16 \mu\text{F}$, $q = 64 \mu\text{C}$)



- (A) $64 \mu\text{J}$ (B) $384 \mu\text{J}$ (C) $320 \mu\text{J}$ (D) $576 \mu\text{J}$
4. A thermally insulated cylinder of volume V contains n moles of ideal monoatomic gas under a weightless piston. A load of weight w is suddenly placed on the piston as a result of which the piston is displaced. If the initial temperature of the gas is 300 K area of piston is A and atmospheric pressure P_0 (such that $P_0 A = w$) Determine the value of final temperature of the gas.
- (A) 420 K (B) 425 K (C) 475 K (D) 515 K

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

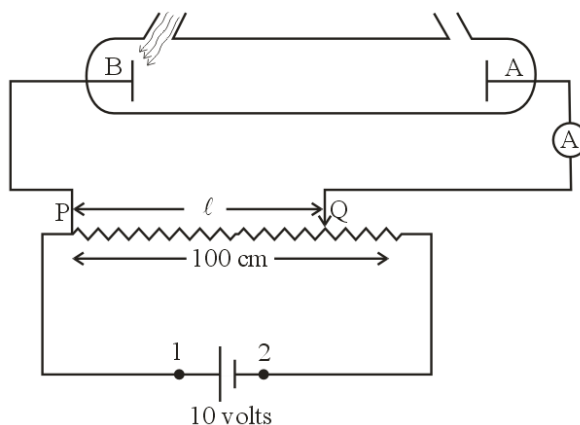
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. In the experiment shown in figure an ammeter is connected in series with an evacuated tube containing two electrodes. The combination is placed in parallel with a potentiometer wire having in series a cell of emf 10 volts. A source of light illuminates the electrode B. The light is composed of four different wavelengths of 155 nm, 310 nm, 620 nm and 1220 nm. the work function of the electrode B is 1.4 eV. On the potentiometer wire jockey is kept at a point Q. Length PQ = ℓ cm.

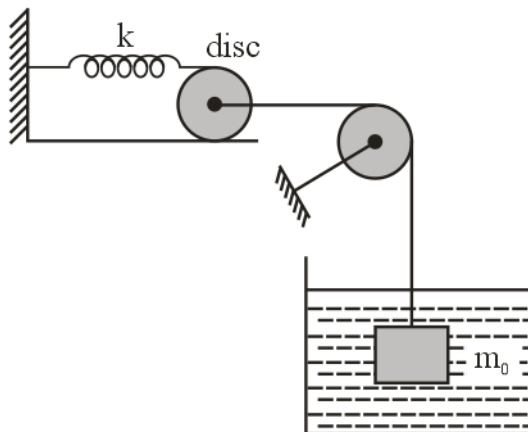


- (A) For $\ell = 60$ cm, the maximum kinetic energy of photoelectron striking electrode A is 0.6 eV
- (B) For $\ell = 80$ cm, ammeter shows no deflection.
- (C) For $\ell = 42$ cm, the deflection in ammeter is more in comparison to that for $\ell = 60$ cm, but the maximum kinetic energy of photo electron striking electrode A is four times of that for $\ell = 60$ cm.
- (D) If the terminals of cell are interchanged, the maximum kinetic energy of striking electron becomes 18 eV.

6. Consider a particle of mass m orbiting in a potential of the form $U(r) = \frac{-k}{r^a}$, where r is the orbital radius and k and a are constants. Applying the conditions of Bohr's model to this particle, it can be said that
- (A) The orbital radius of the particle will be $r_n = \left(\frac{n^2 h^2}{m a k 4 \pi^2} \right)^{\frac{1}{2-a}}$, where $n = 1, 2, 3, \dots$
- (B) The total energy of the particle in the state with orbital radius r_n is $E_n = \frac{n^2 h^2}{8 \pi^2 m r_n^2} - \frac{k}{r_n^a}$.
- (C) The kinetic energy and potential energy are related by $k E_n = -\frac{a}{2} P E_n$.
- (D) The energy of the particle will increase as it moves to orbits of greater radius.

Space for Rough Work

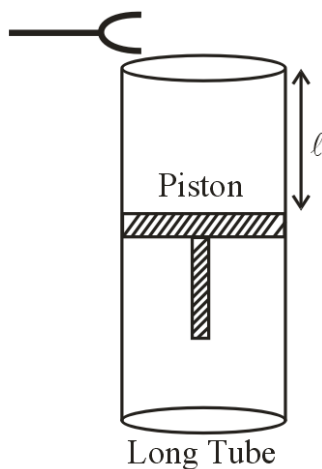
7. A disc of mass 'm' and radius 'R' is kept on rough horizontal surface in a vertical plane. A massless spring of spring constant 'k' is attached at the top of disc, the center of disc is connected with block of mass m_0 which is submerged in water as shown in figure. The relative density of block is 3, there is sufficient friction between disc and the surface to prevent slipping. Take $m_0 = 2m$ and g = acceleration due to gravity.



- (A) The elongation of spring in equilibrium is $\frac{2mg}{3k}$
- (B) The elongation of spring in equilibrium is $\frac{mg}{3k}$
- (C) If disc is slightly displaced from its equilibrium and released then its angular frequency will be $\sqrt{\frac{8k}{3m}}$
- (D) If the disc is slightly displaced from its equilibrium and released then its angular frequency will be $\sqrt{\frac{8k}{7m}}$.

Space for Rough Work

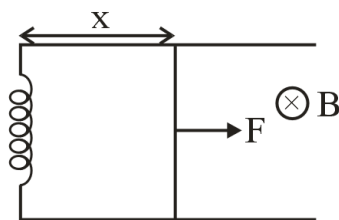
8. A standing wave is formed in a long tube by a tuning fork of frequency 500 Hz kept at its open end and other end is closed by a movable piston as shown in the figure. A resonance is produced. When $\ell = 18$ cm, 55.5 cm and 93 cm respectively. The temperature of gas maintained at 77°C and molecular weight of gas is 33.2×10^{-3} kg/mol choose correct options. (gas constant $R = 8.3\text{ J/mol K}$)



- (A) The minimum distance between node and antinode is 18.75 cm.
 (B) The speed of wave in the tube is approximately 375 m/s
 (C) The adiabatic coefficient of air inside the tube is approximately 1.41
 (D) There is an end correction of 0.75 cm.

Space for Rough Work

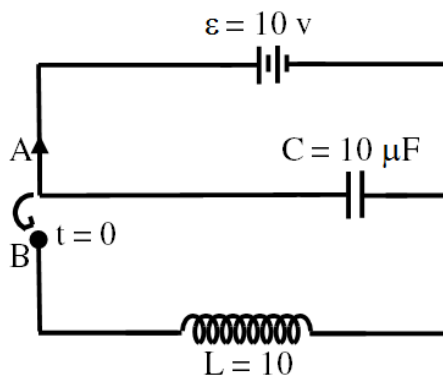
9. A conducting wire of mass 'm' is pulled with uniform velocity v_0 on two parallel rails lying in horizontal plane and separated by distance l . Ends of rails are connected by a coil of self inductance L . At time $t = 0$ $x = x_0 (\cong 0)$ and uniform magnetic field B is present in space. When conducting wire is at distance x from initial position mark correct statement.



- (A) Force F is directly proportional to x
 (B) Force F is directly proportional to x^2
 (C) Energy stored in coil is directly proportional to x
 (D) Energy stored in coil is directly proportional to x^2 .

Space for Rough Work

10. Capacitor of capacitance $C = 10 \mu\text{F}$ was connected with a source of emf $\epsilon = 10 \text{ V}$ for long time. At time $t = 0$ switch S is shifted from point A to B which connects a solenoid of self inductance $L = 10 \text{ H}$, length $\ell = \frac{1}{2} \text{ m}$ and number of turns 10,000. A charge particle of charge $q = 10 \mu\text{C}$, mass $1 \times 10^{-6} \text{ kg}$ is released from rest on the axis of solenoid



- (A) Acceleration of charged particle is zero at $t = 0$
 (B) Acceleration of charged particle is 100 m/s^2 at $t = 0$
 (C) Magnetic field inside the solenoid is nearly $700 \mu\text{T}$ at $t = 10\pi/3 \text{ ms}$
 (D) Magnetic field inside the solenoid is nearly $218 \mu\text{T}$ at $t = 10\pi/3 \text{ ms}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

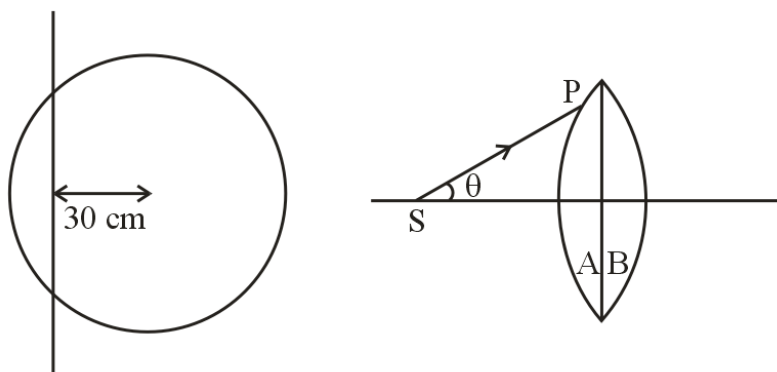
A block of mass 48 kg is attached to the lower end of the rope of mass 24 kg and length 4m. The other end being fixed to the ceiling. A pulse of wavelength $\lambda_0 = \sqrt{\frac{2}{3}} \times 6$ meter is produced at the lower end of the rope. The wavelength of pulse when it reaches the other end of the rope is λ and the speed of pulse at mid point of rope is v (given $g = 10 \text{ m/s}^2$)

1. The value of λ is (in meter)
2. The value of v is (in m/s)

Space for Rough Work

Paragraph for Questions 3 and 4

There are two spheres of transparent materials. Sphere A has refractive index of $\mu = \frac{8}{5}$ and refractive index of sphere B is not known to us (consider it as μ_2). Radii of each sphere is 50 cm. Both the spheres are cut so that the section of cut is 30 cm from center. Later the smaller parts struck together as shown.



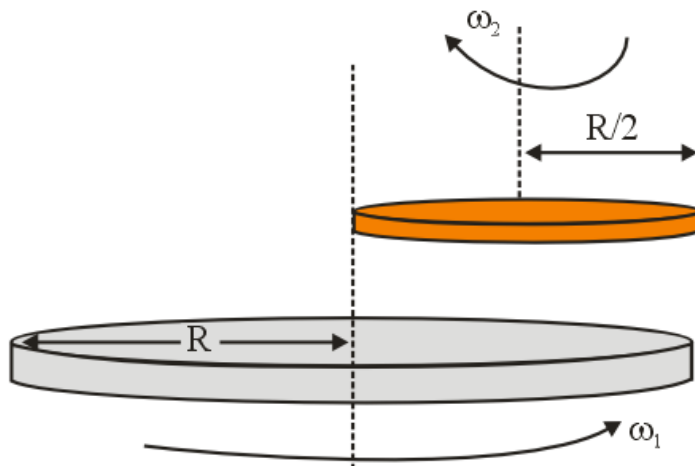
A laser beam from point S (source) is incident on the curved surface of the part of sphere A at point P. The length of beam from S to P is 50 cm. Also the angle made by beam with the principal axis is $\theta = 37^\circ$ ($\sin \theta = \frac{3}{5}$). The beam finally emerges from the curved surface of the part of sphere B. The beam follows a total deviation of 53° .

3. If the image of first refraction is formed at a distance D cm from the surface of A, find the value of $\frac{1}{D}$.
4. If the beam after emerging from B meets the principal axis at I, the total length of beam (in cm) from S to I will be given by $\frac{10}{7}Q$. Find the value of Q.

Space for Rough Work

Paragraph for Questions 5 and 6

A large uniform disk rotates counter clockwise (when viewed from above) with angular speed $\omega_1 = 1 \text{ rad/s}$ about an axis through its center of mass and perpendicular to the flat surface of the disk. A second uniform disk with the same mass 'm' as that of first disk and half the radius, rotates clockwise when viewed from above with angular speed $\omega_2 = 3 \text{ rad/s}$ about an axis through its center of mass and perpendicular to the flat surface of the disk. Suddenly, the second disk is allowed to fall vertically onto the first disk and the two disks finally move with common angular velocity. [$m = 10 \text{ kg}$, $R = 1 \text{ m}$]



5. If the angular velocity of the disks finally after relative motion stops is $\beta \text{ rad/s}$ then find the value of 49β .
6. What should be the average value of torque required so that the first disk keeps on rotating with $\omega_1 = 1 \text{ rad/s}$ during 10 sec of time of relative motion of the two disks ?

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

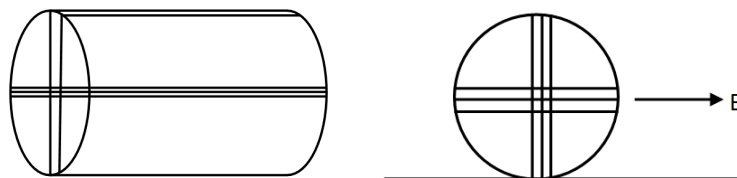
- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A solid cylinder of mass $m = 1$ kg has two sets of identical coils wrapped over it as shown in figure. The plane of the coils are perpendicular to each other. This cylinder is kept on a rough surface. (assume pure rolling)

There is a horizontal uniform magnetic field of strength 1 tesla. The cylinder is released from a position where it has maximum acceleration.

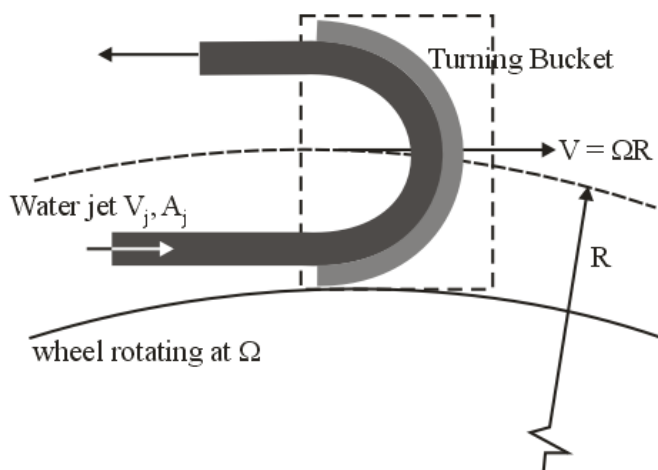


The length of cylinder is 0.1 m, radius $R = \frac{1}{\sqrt{2}}$ m and number of turns in each coil is $N = 10$.

If the maximum velocity acquired by the cylinder be $v = 2$ m/s. The current in the coil in amperes is I_0 . Find the value of $2I_0$ (assume the mass of the coil to be negligible.) (Also ignore effect of induced magnetic field due to coils)

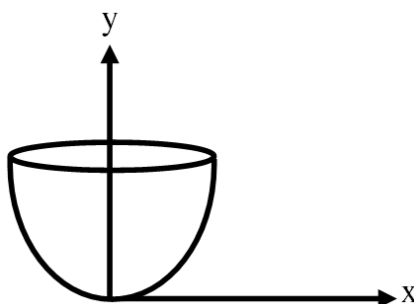
Space for Rough Work

8. A turbine wheel is powered by a water jet as shown in figure.



The turbine is rotating at a constant angular speed. The water jet has a speed of 9 m/s and the radius of wheel is 1 meter. Assuming no loss of speed of water jet relative to turning bucket, find the angular velocity in rad/s at which maximum power will be supplied to the water wheel.

9. A bowl has a shape of a paraboloid as shown in figure. The vertical cross section of which is a parabola.



The radius at the top is 4 m and the maximum depth is 4 m. It is filled completely with a non-viscous liquid. A small hole of cross sectional area $\pi \text{ cm}^2$ made at the bottom. The time required for all the water to flow out from the hole is given by $t = \frac{8}{15} \sqrt{N} \times 10^4$ seconds. Find N.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Two genius students decided to work on a method of preparation of primary amine

KESHAV - Method-1 : $\text{CH}_3 - \text{Br} + \text{NH}_3 \rightarrow$

Method-2 : $\text{CH}_3 - \text{OTs} + \text{NaN}_3 \xrightarrow[\text{Work-up}]{\text{Reductive}}$

GAURANG - Method-1 : $\text{CH}_3 - \text{I} + \text{NH}_3 \rightarrow$

Method-2 : $\text{CH}_3 - \text{I} + \text{NaN}_3 \xrightarrow[\text{Work-up}]{\text{Reductive}}$

Which combination will produce the **BEST** result ?

(A) KESHAV $\rightarrow$ Method-1

(B) KESHAV $\rightarrow$ Method-2

(C) GAURANG $\rightarrow$ Method-1

(D) GAURANG $\rightarrow$ Method-2

Space for Rough Work

2. The **INCORRECT** order of boiling point is
- (A) $\text{H}_2\text{O} > \text{CH}_3\text{OH}$ (B) $\text{N}(\text{CH}_3)_2 < \text{CH}_3\text{OH}$
 (C) $\text{H}_3\text{PO}_4 > \text{Me}_3\text{PO}_4$ (D) $\text{CH}_3\text{N}_3 > \text{HN}_3$
3. In which of the following, the given property varies in the same order as given for different compounds mentioned.
- (A) Lattice energy - $\text{BaO} < \text{CaO} < \text{SrO} < \text{MgO}$
 (B) Thermal stability - $\text{CaCO}_3 < \text{BeCO}_3 < \text{Na}_2\text{CO}_3 < \text{MgCO}_3$
 (C) Solubility - $\text{K}_2\text{SO}_4 > \text{BeSO}_4 > \text{CaSO}_4 > \text{BaSO}_4$
 (D) Hydration energy - $\text{Fe}^{3+} < \text{Al}^{3+} < \text{Co}^{2+} < \text{Fe}^{2+}$
4. Three samples of H_2O_2 which are labeled as 10 vol., 20 vol. and 30 vol. are mixed by taking 100 ml of each and the mixture is diluted to 1000 ml. The molar concentration of the resultant solution is :
 [Consider vol. strength at STP]
- (A) 0.893 M (B) 0.53 M (C) 11.2 M (D) 1.0 M

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

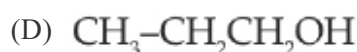
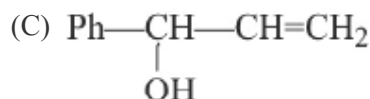
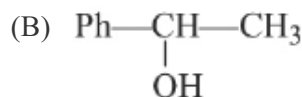
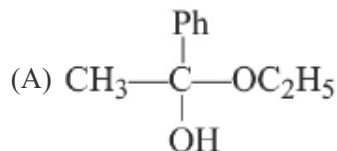
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Which of the following alcohol gives white turbidity immediately with $\text{HCl} + \text{ZnCl}_2$ (anhy.) ?



6. Which of the following metal is not refined by fractional distillation ?

(A) Zn

(B) Cd

(C) Hg

(D) Sn

Space for Rough Work

7. Which of the following is correct for a non-ideal solution of liquids A and B, showing negative deviation from Raoult's law?
- (A) $\Delta H_{\text{mix}} = -ve$ (B) $\Delta V_{\text{mix}} = -ve$ (C) $\Delta S_{\text{surr}} > 0$ (D) $\Delta G_{\text{mix}} = -ve$
8. A_2O_n is oxidized to AO_3^- by MnO_4^- in acid medium. If 1.5×10^{-3} mol of A_2O_n requires 60 ml of 0.03 M $KMnO_4$ solution for complete oxidation, which of the following statement(s) is/are **CORRECT** ?
- (A) The value of $n = 2$
- (B) Empirical formula of oxide is AO
- (C) 0.15 mole of A_2O_n would required 0.15 mole of $K_2Cr_2O_7$ solution
- (D) "A" must be an alkaline earth metal
9.
$$\begin{array}{c} \text{CD}_3 - \text{CH} - \text{CH}_3 \\ | \\ \text{Cl} \end{array} \xrightarrow{\text{alc. KOH}}$$
- True about above reaction is/are :
- (A) It is E2 elimination
- (B) Saytzeff's alkene is major product in above reaction
- (C) $\text{CD}_3 - \text{CH} = \text{CH}_2$ is major product
- (D) It is anti elimination
10. If uncertainty in position of an electron is the same as uncertainty in its momentum, then the uncertainty in position & momentum of electron simultaneously cannot be.
- (A) $\frac{1}{2} \sqrt{\frac{h}{\pi}}$ (B) $\frac{1}{3} \sqrt{\frac{h}{\pi}}$ (C) $\frac{1}{4} \sqrt{\frac{h}{\pi}}$ (D) $\frac{1}{8} \sqrt{\frac{h}{\pi}}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

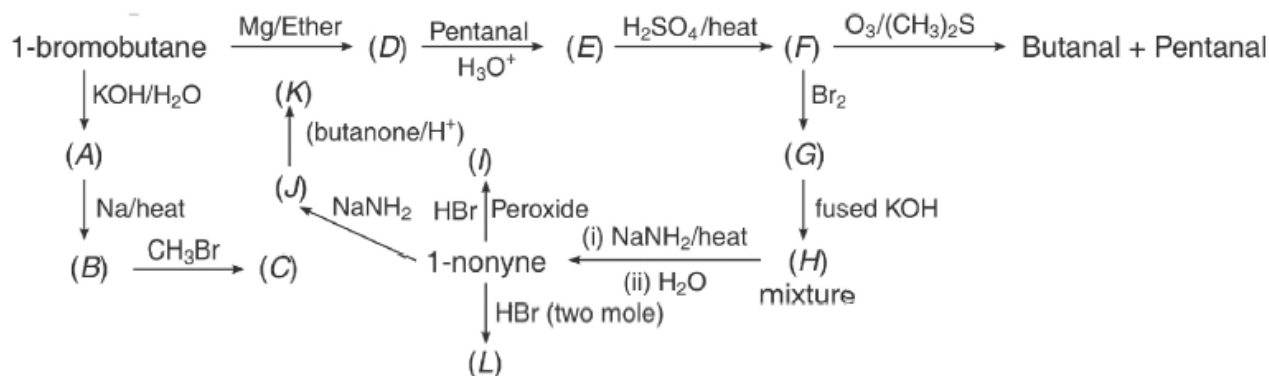
A solution is made originally 0.1 M in Ag^+ ions and 0.25 M in KCN. If dissociation constant of the complex $\text{Ag}(\text{CN})_2^-$ is 3×10^{-19} , what will be the concentration of Ag^+ ions in this solution and what will

be the deposition potential of Ag? [$E^\circ (\text{Ag}^+/\text{Ag}) = 0.8 \text{ V}$, $\frac{2.303RT}{F} = 0.1$, $\log 2 = 0.3$, $\log 3 = 0.48$]

1. $[\text{Ag}^+] = \underline{\hspace{2cm}} \times 10^{-17} \text{ M}$
2. | Deposition Potential of Ag | = volts { || represents mod function }

Space for Rough Work

Paragraph for Questions 3 and 4



3. Find molecular weight of (K) = _____
4. Report molecular weight of (G) = _____

Paragraph for Questions 5 and 6

Polythionic acids have the general formula : $\text{HO}_3\text{S}-\text{S}_n-\text{SO}_3\text{H}$ ($n = 1, 2, 3, \dots$) and their basicity is two. Each 'S' is sp^3 hybridised and the oxidation state of individual 'S' is different. Salts of polythionic acids are named polythionates.

5. The difference in the oxidation numbers of the two types of sulphur atoms in $\text{Na}_2\text{S}_4\text{O}_6$ is _____
6. The highest possible basicity of Phosphorus analogue of $\text{H}_2\text{S}_2\text{O}_6$ will be _____

Space for Rough Work

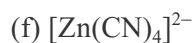
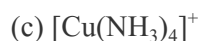
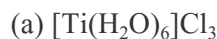
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

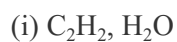
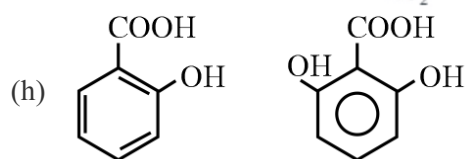
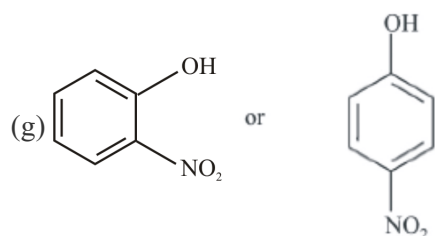
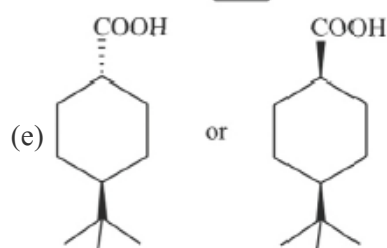
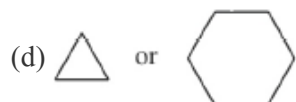
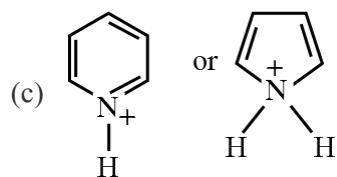
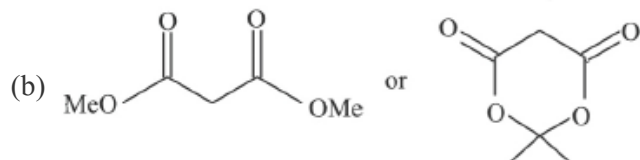
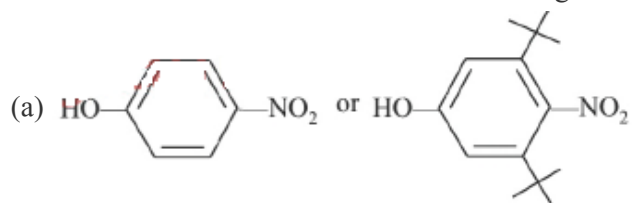
7. Select the complexes which will absorb in visible range :



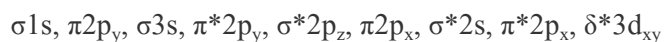
Hence, write total number of such complexes :

Space for Rough Work

8. Write total no. of cases where 1st acid is stronger than then 2nd.



9. How many total number of molecular orbitals which are known as gerade(g).



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $S = \frac{1}{20!} - \frac{1}{19!2!} \times \frac{3}{2} + \frac{1}{18!3!} \times \frac{11}{6} - \frac{1}{17!4!} \times \frac{25}{12} + \dots$ upto 20 terms, then $(21!)S$ is equal to
 (A) 1 (B) $\frac{1}{21}$ (C) -1 (D) $\frac{1}{20}$
2. If $1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \frac{1}{13} - \frac{1}{16} + \dots = \frac{1}{a} \left(\frac{\pi}{\sqrt{b}} + \log_e c \right)$, then the value of $a + b + c$ is
 (A) 6 (B) 5 (C) 8 (D) None of these
3. A rectangle ABCD of dimension 'a' and '2a' is folded along diagonal BD such that planes ABD and CBD are perpendicular to each other. If C point goes to C', then the distance AC' is
 (A) $\sqrt{3} a$ (B) $\sqrt{85} a$ (C) $\frac{\sqrt{85}}{5} a$ (D) $\sqrt{\frac{17}{5}} a$
4. Let ABC is a triangle with incentre I. Let P and Q be the feet of the perpendiculars from A to BI and CI respectively, then $\frac{AP}{BI} + \frac{AQ}{CI}$ is equal to
 (A) $\cot A$ (B) $\cot \frac{A}{2}$ (C) $\tan \frac{A}{2}$ (D) $\tan \frac{B}{2} \tan \frac{C}{2}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Let f be a differentiable function satisfies $\int_0^x ((x+1-t)f(t) - 4t^3 - 2t)dt = 0 \quad \forall x \geq 0$ then the value

of $\int_0^1 \frac{12}{f'(x) + f(x) + 10} dx$ is equal to

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{2}$

Space for Rough Work

6. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three linearly independent vectors and $\vec{d}$ is a non zero vector which is perpendicular to $\vec{a} + \vec{b} + \vec{c}$. Now $\vec{d} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$ then

(A) $\frac{\vec{d} \cdot (\vec{a} + \vec{c})}{\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix}} = 2$

(B) $\frac{\vec{d} \cdot (\vec{a} + \vec{c})}{\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix}} = -2$

(C) Minimum value of $x^2 + y^2 = \frac{\pi^2}{4}$

(D) Minimum value of $x^2 + y^2$ is $\frac{5\pi^2}{4}$

7. Let a, b, c be distinct complex numbers with $|a| = |b| = |c| = 1$ and z_1, z_2 be the roots of the equation $az^2 + bz + c = 0$ with $|z_1| = 1$. Let P and Q represent the complex numbers z_1 and z_2 in the argand plane with $\arg\left(\frac{z_1}{z_2}\right) = \theta$, $0^\circ < \theta < 180^\circ$, then

(A) $b^2 = ac$, $\theta = \frac{2\pi}{3}$

(B) $\theta = \frac{2\pi}{3}$, $PQ = \sqrt{3}$

(C) $PQ = 2\sqrt{3}$; $b^2 = ac$

(D) $\theta = \frac{\pi}{3}$, $b^2 = ac$

Space for Rough Work

8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f(x) + f\left(\frac{x}{4}\right) = x^2 + 2f\left(\frac{x}{2}\right)$ and $f(0) = 1$. Then which of following options is/are correct.
- (A) $f(1) = \frac{25}{9}$ (B) $f\left(\frac{1}{4}\right) = \frac{10}{9}$ (C) $f'(0) = 0$ (D) $f'(1) = \frac{32}{9}$
9. Let $0 < p < q$ and $a \neq 0$ such that the equation $px^2 + 4\lambda xy + qy^2 + 4a(x + y + 1) = 0$ represents a pair of straight lines, then a can lie in the interval (where $\lambda \in \mathbb{R}$)
- (A) $(-\infty, \infty)$ (B) $(-\infty, p]$ (C) $[p, q]$ (D) $[q, \infty)$
10. The number of ways of choosing triplet (x, y, z) such that $z \geq \max\{x, y\}$ and $x, y, z \in \{1, 2, 3, \dots, n, n+1\}$ is
- (A) ${}^{n+3}C_3 + {}^{n+2}C_3$ (B) ${}^{n+2}C_2 + 2 {}^{n+2}C_3$
 (C) $1^2 + 2^2 + \dots + n^2$ (D) $2 {}^{n+2}C_3 - {}^{n+1}C_2$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

In a tournament there are 12 players $P_1, P_2, \dots, P_{12}$. They are divided in 6 pairs at random. From each game a winner is decided on the basis of game played between the two players of the pair. Assume that all players are of equal strength.

1. Probability that exactly one of P_2 and P_7 is among the losers is K then $11K$ is
2. Probability that both P_3 and P_5 are among the winners is λ then 44λ is

Space for Rough Work

Paragraph for Questions 3 and 4

Let $x + y = 2$ and $x - y = 0$ be the tangent at vertex and axis of a parabola respectively straight line $2x - y - 4 = 0$ is a tangent to this parabola.

3. If focus of parabola is (p, q) then $3(p + q)$ is
4. The locus of image of focus about the tangent any point on the parabola is $ax + by - 8 = 0$ then $|a + b|$ is

Paragraph for Questions 5 and 6

Let $\alpha, \beta, \gamma, \delta$ be the roots of the equation $x^4 - 5x^3 + kx^2 - 3x - 1 = 0$ and α, β be the roots of equation $x^2 - ax - 1 = 0$, where as γ, δ be the roots of equation $x^2 - bx + 1 = 0$.

5. The number of real roots of the equation $x^4 - 5x^3 + kx^2 - 3x - 1 = 0$ is
6. The value of $a + b + k$ is

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. Let a, b, c, p, q, r, x, y and z are non-negative real numbers such that $a^7 + b^7 + c^7 \leq 1$, $p^7 + q^7 + r^7 \leq 1$ and $x^7 + y^7 + z^7 \leq 1$, then the maximum value of $ap^2x^4 + bq^2y^4 + cr^2z^4$ is
8. Let a, b, c, x, y and z be complex numbers such that $x = \frac{b+c+2a}{a}$, $y = \frac{c+a+2b}{b}$ and $z = \frac{a+b+2c}{c}$. If $xy + yz + zx = 2011$ and $x + y + z = 2012$, then the value of $|xyz|$ is
9. If $\left(1 + \frac{1}{x}\right)^{x+1} = \left(1 + \frac{1}{2021}\right)^{2021}$, then $|x|$ is equal to
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work